Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Module 2
Clustering

Module 3
Unsupervised Methods for Text Analysis

Lesson 1: Text Representations

How to Represent Text as a Vector: A Typical Workflow
Video • 21 min

Time to Practice: Representing Text as a Vector
Graded Assignment • Grade: 100%

Text Processing in SciKit-Learn
Video • 13 min

Time to Practice: Text Processing in SciKit-Learn
Graded Assignment • Grade: 100%

Lesson 2: Topic Modeling Methods & Techniques

Introduction to Topic Modeling
Video • 18 min

Latent Dirichlet Allocation (LDA)
Video • 7 min

Using LDA with Scikit-Learn
Video • 11 min

Time to Practice: Latent Dirichlet Allocation (LDA)
Graded Assignment • Grade: 100%

Non-Negative Matrix Factorization (NMF)
Video • 19 min

Time to Practice: Non-Negative Matrix Factorization (NMF)
Graded Assignment • 15 min

Lesson 3: Word Embeddings Techniques

Word Embeddings Technique #1: Word2vec
Video • 21 min

Word Embeddings Technique #2: Glove
Video • 16 min

Time to Practice: Word2vec & Glove
Graded Assignment • 15 min

Module 3 Wrap-Up

Introduction to Module 3 Assignment: Text Representations, Topic Modeling, and Word Embeddings
Reading • 5 min

Create & Submit Module 3 Assignment
Programming Assignment • Grade: 100%

Assignment 3 Followup
Dialogue • 15 min

Module 3 Optional Readings & Resources
Reading • 5 min

Module 3 Optional Readings & Resources

0:15 / 0:35 1.5x

- **Textbook Chapter:** Introduction to Machine Learning with Python Chapter 7: Working with Text (all sections)
- **Article:** Probabilistic Topic Models For a bit more depth into topic models and specifically Latent Dirichlet Allocation (LDA), this ACM survey article by David Blei, the principal creator of LDA topic modeling, is an accessible summary of how it works and how it can be applied.
- **Article:** Lego Color Themes as Topic Models This article by Nathanael Aff demonstrates how the concepts and algorithms of topic modeling can be applied to more than just words and documents!

Go to next item ✓ Completed

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